

AMERICAN INSTITUTE OF MATHEMATICS

WORKSHOP FINAL REPORT Addressing Declining Pollinator Populations Through New Mathematics

March 30 – April 5, 2026

American Institute of Mathematics (AIM) • <https://aimath.org/pastworkshops/year2026>

Workshop Mission

This workshop united mathematicians, ecologists, data scientists, and beekeeping practitioners to develop rigorous, actionable mathematical frameworks capable of understanding, predicting, and ultimately reversing the decline of honeybee and pollinator populations. The challenge is urgent: bees pollinate approximately 80% of the world’s food crops and contribute over \$14 billion annually to U.S. agriculture — yet annual colony losses now routinely exceed 30–50%, a level that threatens global food security and ecological stability.

Executive Summary

The AIM workshop on “Addressing Declining Pollinator Populations Through New Mathematics” convened a uniquely broad interdisciplinary community — mathematicians, biologists, data scientists, and commercial beekeeping practitioners — over five days in March–April 2026. The workshop’s central finding is clear: pollinator decline is a complex adaptive systems problem that demands mathematical sophistication, sustained cross-disciplinary collaboration, and direct engagement with the practitioners who can translate research into action. This report synthesizes all eight working group outcomes and presents a coordinated 12-month research and partnership roadmap.

The Core Challenge

Colony collapse cannot be explained by any single stressor. Varroa mites, viral infections, nutritional deficits, pesticide exposure, queen quality, and climate stress particularly heat waves, drought, and delays in cooler fall temperatures, interact synergistically in ways that produce sudden, nonlinear tipping points. No model, dataset, or discipline can address the complexities of colony collapse —

which is precisely why this interdisciplinary, multi-sector workshop matters.

Key Findings at a Glance

- Pollinator decline is a multi-stressor, multi-scale systems problem. It requires that integrated mathematical, ecological, data-driven, and practitioner-engaged approaches occur simultaneously.
- Seven critical stressor categories were identified across all groups: Varroa destructor mites; nutritional quality and landscape forage; climate variability and extreme weather; viral infections (multiple viruses, not only deformed wing virus); pesticide and agrochemical exposure; cumulative forager flight hours; and queen health and colony demography.
- The most powerful mathematical opportunities are: threshold and resilience analysis; stochastic and nonautonomous dynamical systems; optimal intervention design; early warning signal detection; geospatial-landscape integration; and hybrid mechanistic–machine learning modeling.
- Existing mathematical results from workshop participants — Allee effects, catastrophic events, multi-attractor dynamics, Hopf bifurcations in honeybee-mite systems — already provide a rigorous theoretical foundation. The workshop defined exactly how to extend and connect these results into an integrated research program.
- No single dataset currently captures all seven stressors across multiple colonies over a full annual cycle. Addressing this data infrastructure gap is the field’s most urgent near-term priority.
- The workshop produced a concrete coordinated 12-month plan built around shared data infrastructure, pilot manuscripts, and interdisciplinary grant development that spans institutions, sectors, and disciplines.

01

Why This Work Is Urgent

The ecological, economic, and mathematical case for sustained investment

Honeybees are the linchpin of global agriculture and terrestrial ecology. They pollinate nearly 80% of the world’s food crops, contribute over \$14 billion annually to U.S. seed and crop production, and sustain the wildflower diversity on which thousands of other species depend. The accelerating collapse of managed and wild bee populations is not merely an agricultural inconvenience — it is a slow-motion ecological emergency with direct consequences for food security, biodiversity, and human welfare.

Annual U.S. honeybee colony losses have exceeded 30–50% in recent years, against a historical baseline of approximately 10%. These losses have persisted despite decades of research and intensive management, strongly suggesting that the problem is fundamentally more complex than any single-factor or simple explanation can capture. Finding solutions to colony losses will require the identification of interactions among multiple factors that set a colony on a course for collapse. Mathematics is uniquely positioned to identify those interactions and provide timely recommendations for interventions that can reverse the course and prevent colony loss.

Mathematics offers an essential lens to examine colony losses. Prior work by workshop participants — including foundational models of honeybee-mite interactions, virus transmission dynamics, dispersal effects, age structure, and nutritional impacts — has already demonstrated that mathematical modeling can identify hidden thresholds, predict collapse trajectories, and guide management timing. The workshop’s goal was to accelerate and broaden that program, connecting individual modeling efforts into a coherent, collaborative research enterprise.

Historical Perspective

Concerns about bee decline predate Colony Collapse Disorder by nearly a century. Entomologist Edith Patch warned in the 1930s about declining bee populations driven by agricultural intensification, reduced forage diversity, and early pesticide use. The problem has deep roots — requiring sustained, systemic attention and the kind of long-term interdisciplinary research program this workshop has launched.

02

Scientific Synthesis Across Working Groups

Integrated findings from all eight thematic groups

Eight working groups addressed distinct but deeply interconnected dimensions of the pollinator decline problem. The findings below reflect both group-specific results and the cross-cutting connections that emerged during joint sessions.

2.1 Colony Persistence, Collapse, and Tipping Points (Group 1)

Group 1 established the theoretical core of the workshop. Building on the Perry-Sovik-Myerscough-Barron framework (PNAS 2015) and prior honeybee-mite-virus models developed by workshop participants, the group focused on the mathematical conditions under which a colony persists, oscillates, or is irreversibly driven to collapse.

The central mathematical concept is the Allee effect: honeybee colonies require a critical minimum population to sustain themselves, and below that threshold, collapse is inevitable regardless of external conditions. This produces a bistable dynamical system with two stable equilibria — a healthy colony and extinction — separated by a tipping point. Critically, collapse can be sudden and path-dependent: how a colony arrives at a given population level matters as much as the level itself.

A key conceptual contribution was the proposal of a composite “stress score” that aggregates multiple stressor effects into a tractable parameter for initial modeling, enabling staged model development without sacrificing biological realism. Jordan Rodriguez (UT Arlington) introduced a colony health analysis method adapted from soil science, offering a promising quantitative framework for measuring overall colony condition across datasets.

Key

Synergy drives collapse more than any individual stressor

Insight

Even within apiaries with genetically similar queens, comparable mite treatments, and shared forage environments, colony outcomes diverge dramatically. This unexplained variance is the signature of hidden synergistic dynamics — the very dynamics that mathematical modeling is uniquely equipped to reveal, characterize, and ultimately predict.

2.2 Optimization of Beekeeping Management Interventions (Group 2)

Group 2 translated the abstract concern about stressor interactions into a set of mathematically tractable, operationally relevant optimization problems. The central insight was that optimal management timing is not a fixed calendar rule but a state-dependent decision that depends on current colony trajectory, mite loads, brood levels, and seasonal dynamics.

Two parameters emerged as especially tractable for near-term optimization with existing data:

- Miticide application timing:
 - The current standard threshold (1 mite per 300 bees) is a treatment trigger, not an optimum. ODE-based optimization models can identify treatment windows as a function of brood-to-adult ratios, mite infestation rates, and colony trajectory rather than static counts — potentially preventing colony losses that current calendar-based approaches miss.
 - Colony split size and timing: Beekeepers commonly divide a large colony into smaller ones called ‘splits’. The group proposed a field experiment to test whether a critical viable split size exists. The hypothesis: 2-frame splits likely cannot sustain themselves; 5-frame splits reliably succeed; and the 3–4 frame range is the key decision zone. Biweekly monitoring of splits created at different times would generate data directly usable for model validation.

Additional intervention parameters identified for future optimization include timing for supplemental feeding, nutritional composition of feeding supplements, , queen replacement scheduling, drone brood removal, antibiotic and fungicide application, and overwintering preparation strategies. Each represents a point where mathematical optimization can improve on current rule-of-thumb practice.

2.3 Climate and Agrochemical Effects in Mechanistic Models (Group 3)

Group 3 addressed one of the most consequential gaps in current colony models: the explicit representation of climate forcing and agrochemical exposure. The group reached consensus that stochastic, nonautonomous frameworks are required to capture the biological realities of temperature variability, extreme weather events, and dose-dependent chemical effects.

Priority modeling directions:

- Temperature-dependent population dynamics that respond to both mean conditions and the frequency and magnitude of extremes such as heat waves. Egg-laying rates, brood survival, and forager performance are all thermally sensitive in nonlinear ways.
- Energetic costs of thermoregulation under climate stress, that divert colony resources from brood rearing and foraging and may synergistically amplify nutritional deficits.

- Phenological mismatch modeling: climate change is shifting bloom timing in ways that may decouple the availability of floral resources from periods of peak brood rearing, with consequences that are not captured by current static nutrition models.
- Toxicity functions for agrochemicals, including sublethal behavioral effects, immune suppression, and synergistic interactions with mite loads and viral infections that multiply the damage of any single exposure.

The group has committed to a joint paper integrating biological insights and mathematical modeling of climate impacts on honey bees — one of the workshop’s first concrete publication targets, and a model for the kind of cross-disciplinary synthesis the workshop was designed to produce.

2.4 Early Warning Indicators of Colony Decline (Group 4)

Group 4 identified one of the workshop’s most immediately actionable research directions: using continuous sensor data and behavioral monitoring to detect the precursors of collapse before it becomes irreversible. This is a domain where mathematics and data science can make an immediate, practical difference.

Julia Fine (USDA ARS) presented a landmark dataset: one year of 15-minute-resolution data from 40 colonies using in-hive sensors measuring internal acoustics, temperature, humidity, and external apiary conditions, paired with monthly manual population and brood counts. This dataset is unprecedented in its combination of spatial, temporal, and biological coverage and will serve as a key validation resource for early warning models.

The group’s analytical roadmap:

- Brood-to-adult ratios and mite-to-bee ratios as leading indicators: periods of low brood relative to adults signal both optimal oxalic acid treatment windows and potential early collapse trajectories.
- Acoustic and thermal signatures: changes in hive acoustics and temperature patterns may precede detectable demographic decline by days to weeks, enabling proactive rather than reactive intervention.
- Time series analysis, correlation structure, clustering, and predictive modeling to identify pre-collapse signatures in high-frequency sensor streams.
- Economic integration: any intervention decision tool must account for the cost of false alarms as well as missed detections and must be actionable on a commercial scale without specialized expertise.

Commercial Imperative	Models must translate into operational decisions that beekeepers can actually use
	Beth Conrey (B Squared Apiaries, Colorado) made clear that commercial beekeepers need decision rules, not diagnostic dashboards. An early warning system that requires a PhD to interpret will not prevent colony losses. Mathematical simplification for practical deployment is not a compromise — it is a research goal in its own right.

2.5 Geospatial and Landscape-Integrated Models (Group 5)

Group 5 reframed landscape nutrition from a background variable to a foundational driver of colony health. The quality, diversity, and temporal availability of floral resources in the surrounding landscape profoundly shape colony outcomes, even when supplemental feeding is provided — a finding with direct implications for agricultural land management, conservation policy, and the design of any colony dynamics model that aspires to predictive accuracy.

The group’s methodological agenda drew on deep learning, citizen science data, and geospatial platforms to build the landscape input layer that colony models currently lack:

- High-resolution aerial imagery analyzed with deep learning models to predict flowering species richness across diverse landscapes.
- Phenological curves inferred from citizen science temporal data to map when and where floral resources become available throughout the season.
- Image segmentation applied to landscape photographs to quantify floral abundance at fine spatial scales.
- Machine learning analysis of bee waggle dance patterns to understand colony-level foraging decisions and actual landscape use in real time.
- Cross-referencing of foraging maps with pesticide application records to estimate actual field-level chemical exposure and identify risk mitigation strategies.

2.6 Machine Learning with Multi-Scale Data (Groups 7 & 8)

Groups 7 and 8 addressed the methodological frontier of hybrid mechanistic–machine learning modeling. Pollinator datasets are typically sparse, noisy, heterogeneous, and collected on multiple time scales that do not align. Traditional statistical methods lose power in these conditions; purely data-driven approaches lose biological interpretability. The solution is integration of mechanistic structure with adaptive learning.

Key methodological directions:

- Neural ODEs: embedding known biological structure (the Allee effect, mite reproduction dynamics, age-structured transitions) into neural network architectures that combine mechanistic interpretability with data-adaptive flexibility.
- Data assimilation: updating model state estimates in real time as new sensor or monitoring data arrive, enabling dynamic rather than static prediction of colony trajectories.
- Latent-state estimation: inferring hidden variables — queen condition, cumulative stress, internal viral load — from observable population and behavioral data.
- Uncertainty quantification: producing probabilistic forecasts rather than point estimates, enabling risk-based decision-making that accounts for the genuine unpredictability of complex biological systems.

A Key Methodological Principle

Start with the simplest model that captures the essential biology, validate it against data, then extend modularly. Prior work has shown that simple honeybee-mite models without full age structure can

reproduce qualitative collapse dynamics observed in the field. Adding complexity is only justified when the simpler model fails to account for observed phenomena. This discipline keeps models analytically tractable and biologically interpretable.

03

Cross-Cutting Themes

Six findings that transcended individual working groups

Six themes emerged with striking consistency across all working groups. These represent the workshop's strongest scientific consensus and the most important targets for ongoing interdisciplinary and multi-sector collaboration.

Theme 1

Synergy is the core scientific problem — and mathematics is the right tool for it

Colony collapse arises from the nonlinear, time-dependent interaction of multiple stressors that together push colonies past tipping points no individual stressor could reach. Bifurcation theory, multi-attractor dynamics, and bistability analysis are uniquely suited to characterizing these interactions. No other discipline has equivalent tools. This is why mathematics must be at the center of pollinator research, not at its periphery.

Theme 2

Resilience before collapse: the early warning opportunity

Colonies exhibit measurable signs of stress — in acoustics, thermal gradients, foraging behavior, brood-to-adult ratios — before population collapse becomes observable. Detecting these signals early and translating them into actionable decision rules could prevent enormous losses without requiring new treatments. This is a domain where mathematics can deliver impact on a short timescale.

Theme 3

Landscape nutrition is a hidden driver that current models systematically underestimate

Pollen diversity, sterol content, bloom phenology, and spatial foraging range all shape colony health in ways that supplemental feeding cannot fully compensate for. Models that treat landscape as a static background constant are missing a major source of the variance they are trying to explain. Integrating geospatial forage models with colony dynamics is not optional — it is necessary for predictive accuracy.

Theme 4

Data integration is more urgent than data collection

Multiple high-quality datasets exist across participating labs. What is missing is not data per se but the infrastructure and analytical frameworks to integrate them. Machine learning, neural ODEs, and data assimilation can extract substantial value from existing,

heterogeneous, and incomplete datasets — without waiting a decade for the ideal prospective study.

Theme 5

Translation to practice is a research goal, not an afterthought

Mathematical models will only matter if they produce tools that beekeepers, land managers, and policymakers can use. This requires collaboration with commercial operators from the start, attention to economic and operational constraints, and the development of decision rules robust enough to apply without specialized expertise. The workshop made clear that this translation work must be designed into the research program, not bolted on at the end.

Theme 6

Multi-sector, multi-institutional collaboration is the key multiplier

No single lab, institution, or discipline can solve this problem. The workshop’s most important product may be the collaborative network it created: mathematicians and biologists, academic researchers and commercial beekeepers, theoretical modelers and sensor engineers, university groups and USDA scientists. Sustaining and expanding these relationships — across sectors, institutions, and national borders — is as important as any specific research outcome.

04

Mathematical Foundations

Prior results that anchor the research program

The workshop did not begin from scratch. A substantial body of rigorous mathematical results — developed primarily by workshop participants over the past decade — provides the theoretical foundation on which all new work will build. These results also demonstrate, concretely, that mathematics can contribute unique insights to this problem that no other discipline can provide.

Allee Effects and Catastrophic Events

Honeybee colonies exhibit a strong Allee effect: below a critical population threshold, the colony cannot sustain itself regardless of external resource conditions. Mathematical models show bistability between an extinction equilibrium and a healthy colony equilibrium. When mite infestations push the coexistence equilibrium below a critical carrying capacity threshold, the system undergoes a subcritical Hopf bifurcation: unstable periodic orbits expand until they touch the stable manifold of the boundary equilibrium, driving both honeybees and mites to extinction. This “catastrophic event” is a rigorous mathematical analog of Colony Collapse Disorder and provides a mechanistic explanation for why CCD can appear sudden and irreversible.

Parasite-Virus Synergy and Multiple Attractors

Models incorporating both Varroa mites and viral infections generate qualitatively different dynamics from single-stressor models. The synergistic combination creates multiple attractors and makes disease persistence more likely even in parameter regimes where the individual stressors alone would predict healthy colony survival. This result directly motivates the need for integrated multi-stressor models and explains why treating mites without addressing viral load, or vice versa, may be insufficient.

Dispersal, Patch Dynamics, and CCD Propagation

Two-patch models of mite dispersal via forager bees show that intermediate dispersal rates between the source colony for dispersing mites and the receiving colony where mites immigrate. Large dispersal rates can destabilize both patches simultaneously and drive CCD-like extinction across an entire apiary. This has direct practical implications for apiary layout, management of forager drift, and understanding how collapsing colonies accelerate decline in neighboring healthy ones.

Delay Differential Equations and Staged Modeling

The developmental delay from egg to adult bee introduces time lags that affect how mite parasitism destabilizes colony dynamics. Remarkably, analysis shows that simpler delay-free models can reproduce the qualitative dynamics of fully age-structured systems - justifying a staged modeling strategy that begins with analytically tractable systems and extends complexity only when the simpler model fails to capture observed phenomena.

Optimal Control and State-Dependent Intervention

Mathematical analysis confirms that optimal treatment windows are state-dependent, not calendar-based. The brood-to-adult ratio at treatment time determines both the efficacy of oxalic acid and the colony’s subsequent recovery trajectory. This rigorous result provides direct mathematical justification for the sensor-based early warning tools proposed by Group 4 and the optimization models proposed by Group 2.

05

Priority Research Agenda

The six most important directions for the next 1–3 years

Priority Area	Specific Objective and Scientific Rationale
Baseline colony dynamics model	Develop a healthy-colony population framework anchored in the Perry-Barron ODE system, with modular extensions for mites, viruses, nutrition, pesticides, and climate. This shared mathematical reference will anchor all sub-team work and ensure analytical coherence across the collaboration.
Multi-stressor resilience architecture	Formalize how cumulative and interacting stressors move colonies toward resilience loss and collapse. Identify bifurcation parameters, tipping thresholds, and early warning

Priority Area	Specific Objective and Scientific Rationale
	signatures for each stressor combination. This is the mathematical core of the research program.
Early warning analytics pipeline	Analyze data from in-hive sensors and manual colony data to detect predictive signatures of brood stress, optimal treatment windows, and collapse risk. Time series, machine learning, and optimal control methods will be deployed. Near-term impact potential is high.
Intervention optimization framework	Build optimization or optimal control models for miticide timing, split size, supplemental feeding, and overwintering. Incorporate both colony health and economic outcomes. Validated against field data from commercial operations.
Climate-landscape integration	Link stochastic temperature models, thermoregulatory energy costs, forage phenology, and pesticide exposure landscapes to colony population dynamics in a spatially explicit framework. Capturing the interactions among these variables is essential for predictive validity in the face of climate change.
Hybrid mechanistic-AI methods	Develop neural ODE, data assimilation, and latent-state estimation approaches that can extract biological insight from sparse, heterogeneous, multi-scale pollinator datasets. The methodological frontier that will determine long-term model power.

06

Coordinated 12-Month Future Plan
From workshop outcomes to publications and funded research

The following plan reflects a collective commitment across all working groups. It is designed to maintain momentum, produce early visible outputs, and position the collaboration for competitive external funding within 12 months. Each phase builds on the previous one; delays in Phase I will cascade.

Phase I: Organize and Consolidate (Months 0–3)

Phase I Goal
Build the shared infrastructure — data, code, communication, and sub-team structure — that makes everything else possible. Phase I outputs may be invisible externally but are essential to research program success.

- Create public website through Arizona State University and shared cloud infrastructure (folder, repository, communication channel) for data, references, model code, and documentation accessible

to all participants. The website will be open to public. Ideally, the website will provide links to other resources such as Beescape, Beespatial, and BEEPOP models.

- Gloria DeGrandi-Hoffman shares multi-year colony measurement data (frame counts, Varroa levels, pollen data, virus data) organized with corresponding publications via shared Dropbox.
- Julia Fine (USDA ARS) evaluates feasibility of sharing the in-hive sensor dataset with the broader group and coordinates with Pierre Lau on data sharing logistics.
- Yun Kang and the Group 1 team produce a summary of all available datasets: variables measured, temporal resolution, colony counts, geographic coverage, and format.
- Define sub-teams around the six priority areas with leads, communication schedules, and milestones.
- Conduct power analysis to determine appropriate sample sizes and measurement frequency for a future large-scale monitoring study.
- Schedule monthly focused virtual sessions per sub-team to maintain momentum between in-person gatherings.

Phase II: Pilot Studies and Early Manuscripts (Months 3–6)

Phase II Goal

Produce visible early outputs that demonstrate the collaboration is working, attract additional collaborators, and provide the preliminary results needed for competitive grant applications. Momentum is as important as perfection at this stage.

- Workshop perspective/synthesis article: Define the mathematical agenda for pollinator research, situate it in complex systems biology, and articulate the case for sustained interdisciplinary investment. Target a broad-audience journal (PNAS, PLOS Biology, or Notices of the AMS).
- Pilot early warning study: Apply time series, correlation, and clustering analyses to the in-hive sensor dataset to identify acoustic and thermal precursors to colony decline. Target a journal at the intersection of applied mathematics and entomology.
- Climate sensitivity paper: Draft the Group 3 paper on temperature-dependent and stochastic colony models incorporating extreme weather event impacts.
- Preliminary optimization paper: Develop and analyze an ODE-based optimal control model for miticide timing or split-size decisions, validated against field data from B Squared Apiaries and USDA colony datasets.
- Baseline colony model technical report: Produce a shared mathematical reference document describing the colony dynamics framework with modular stressor extensions.

Phase III: Integrated Grant Development (Months 6–12)

Phase III Goal

Convert pilot results and workshop consensus into competitive, externally funded research programs. Coordinated multi-institution proposals will be stronger than individual submissions and will signal to funders that a genuine research community has formed around this problem.

- NSF Mathematical Biology (DMS) proposal: Multi-stressor mathematical modeling of pollinator resilience. Build on the NSF-DMS awards previously supporting Kang and collaborators (Award Numbers 1313312, 1716802, 2052820). Target the next DMS-MathBio competition.
- USDA-NIFA-AFRI proposal: Large-scale colony monitoring study capturing all seven stressor categories across multiple sites and years. Lead PIs: Fine/Adrian and Kang with both mathematical/data science and biologists.
- NSF-IOS/DMS integrative proposal: Linking colony-level biological dynamics with geospatial landscape models. Target the NSF Integrative Organismal Systems / Mathematical Biology joint mechanism.
- Beekeeping industry formal partnerships: Establish data-sharing and co-development agreements with commercial operations to ensure field validation and translation pathways for all modeling work. This is both a scientific and a funding strategy.
- Additional mechanism exploration: DOE, NIH, or private foundation opportunities for hybrid mechanistic-AI methods applied to ecological forecasting.

07

Recommended Deliverables

Concrete outputs that will define the workshop’s lasting impact

Deliverable	Lead(s)	Timeline	Type
Workshop perspective article-review article	Kang + all PIs	6 months	Paper
Pilot early warning study	Fine, Udiani, Rodriguez	6 months	Paper
Climate sensitivity paper	Group 3 team	9 months	Paper
Intervention optimization paper	Group 2 team	9 months	Paper
Shared dataset crosswalk/inventory	DeGrandi-Hoffman, Kang, Michelle,	3 months	Infrastructure
Baseline colony model (technical report)	Kang, Parshad, Pal	6 months	Tech report
Early warning pipeline (software)	Group 4 team	12 months	Software
Beescape deep learning extension	Kevin Li, Group 5	12 months	Software
NSF MathBio proposal	Kang + collaborators	12 months	Grant
USDA-NIFA-AFRI proposal	Both Math and Bio	12 months	Grant
Commercial beekeeper decision tool	Groups 2 & 4 + industry	18–24 months	Applied tool

08

Workshop Participants

The interdisciplinary, multi-sector community driving this work

The workshop drew on expertise spanning mathematical biology, dynamical systems, ecology, entomology, virology, data science, geospatial analysis, and commercial beekeeping across multiple institutions and countries. This breadth — from differential equations to hive management — is itself one of the collaboration’s most important assets.

Participant	Affiliation	Group(s)	Expertise
Yun Kang	Arizona State University	1,3,4,7	Math biology, honeybee modeling
Gloria DeGrandi-Hoffman	USDA ARS/ Arizona State University	1	Honeybee biology, data, population modelling,
Michelle Flenniken	Montana State University	1	Virology, Bee Biology
Rana Parshad	Iowa State University	1	Dynamical systems
Samares Pal	Univ. of Kalyani	1	Math biology
Vardayani Ratti	CSU Chico	1	Honeybee-virus modeling
Kevin Li	Penn State University	1, 5	Landscape ecology
Jordan Rodriguez	UT Arlington	1, 4	Data analysis, ODEs
Harland Patch	Penn State University	1	Landscape nutrition
Lucero Rodriguez	Workshop Participant	1	Math biology, HIV
Julia Fine	USDA ARS	4	Sensor data, monitoring. Bee biology.
Beth Conrey	B Squared Apiaries (CO)	4	Commercial beekeeping
Pierre Lau	USDA ARS	4	Bee health, toxicology
Oyita Udiani	Virginia Commonwealth Univ.	4	Self-organized systems, Math Bio
John Stavroulakis	Workshop Participant	4	Mathematical modeling, DDE
Oluwatosin Babasola	Workshop Participant	3	mathematical modeling
Kayode Oshinubi	Workshop Participant	3	Mathematical modeling
Oluwaseun Ajayi	Workshop Participant	3	Applied mathematics
Jianzhong Su	UT Arlington	3, 5	Differential equations, dynamical systems

Participant	Affiliation	Group(s)	Expertise
Jun Chen	Texas A&M University - Kingsville	3, 7	ML + dynamical systems
Adrian Fisher II	ASU	3, 7	Heat Waves, Queen egg, laying rates, bee biology/physiology
Hermann J Eberl	Professor, Dept. Mathematics and Statistics		Applied Mathematics, Bee modeling
Hayoung Choi	Department of Mathematics Kyungpook National University	5, 7	Data analysis, AI

09

Closing Statement

Why this collaboration is positioned to make a difference

Honeybees have survived for tens of millions of years. Their current crisis is human-made — a product of industrial agriculture, habitat loss, climate disruption, and the inadvertent introduction of novel pathogens. It is therefore a problem that human knowledge, thoughtfully applied, can address. This workshop was a demonstration that the knowledge exists and the community is forming.

Three things set this collaboration apart from prior efforts in this space:

- **Mathematical depth:** The bifurcation theory (tipping points), delay differential equations, stochastic analysis, optimal control, and machine learning methods available to this group substantially exceed what previous honeybee models have employed. There are genuine mathematical discoveries still to be made in this domain — results that will matter both for pollinator science and for the broader mathematics of complex adaptive systems.
- **Cross-sector breadth:** From differential equations to waggle dance analysis to commercial apiary management, the range of expertise in this collaboration is genuinely unusual and genuinely necessary. That breadth is a multiplier for every individual contribution.
- **Shared urgency:** Participants share the conviction that this problem matters — that a failure to act is not a neutral scientific stance but a choice with real ecological and economic consequences. That shared urgency will sustain the collaboration through the difficult phases ahead.

The workshop’s ultimate measure of success is not publications or grant awards. It is whether honeybee colonies — and the pollinators that share their fate — are healthier a decade from now because of the mathematics done here. That is an ambitious goal. It is also the right one.

A Final Word

“If the bee disappears from the surface of the earth, man would have no more than four years to live. No more bees, no more pollination, no more plants, no more animals, no more man.” — Attributed to

Albert Einstein. Whether or not Einstein said this exactly, the sentiment captures the ecological reality with precision. This workshop, and the research program it has launched, is a concrete step toward ensuring that reality never has to be tested.

Selected References

Some Key works cited or discussed during the workshop:

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